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ABSTRACT

Controlling and understanding thermal energy at the nanoscale is a key challenge in nanoscience, with broad implications for semiconductor technology, photocatalysis, and light-activated medicine. Plasmonic nanostructures offer a powerful approach to achieve sub-wavelength photothermal confinement by converting light into localized heat. In this review, we first examine the fundamental mechanisms of plasmonic light-to-heat conversion and how both classical and non-Fourier heat transfer models describe the spatiotemporal evolution of temperature. We then explore how advances in material science, nanophotonic architectures, and micro-environmental engineering have opened up new opportunities for achieving faster and more localized thermal gradients. Finally, we briefly assess the capabilities and limitations of current nanothermometry techniques for resolving temperature with nanometer precision and at ultrafast timescales.

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INTRODUCTION

Controlling energy at the nanoscale, particularly in the form of heat, represents one of the most promising and intriguing frontiers in modern nanoscience.^{1,2} The precise control of heat transfer enables innovative applications across numerous scientific and technological fields [Fig. 1(a)]. For example, the precise management of nanoscale thermal transport is essential to warrant the performance of ever-shrinking semiconductor devices.^{3,4} Indeed, as device features become smaller, local power densities can easily surpass the heat-flux capacity of conventional thin-film interfaces commonly found in chip technologies, calling for novel approaches for rapid rerouting and removal of heat at the nanoscale. Second, nanoscale heat management can drastically reduce thermal losses, prolonging device lifetimes, enhancing efficiency, and even enabling innovative energy harvesting strategies in photovoltaic technologies.^{5–7} Third, localized heating during photocatalysis can dramatically accelerate catalytic reactions, allowing selective chemical transformations that are otherwise inaccessible or energetically inefficient.⁸ Finally, precise control over nanoscale heat gradients is crucial for hyperthermia therapy and targeted drug delivery in the medical field, particularly so for the selective ablation of tumor tissue.^{9,10}

One interesting platform to investigate the spatiotemporal localization of nanoscale heat is offered by photothermal materials, i.e., materials that efficiently absorb light and convert the absorbed photon energy into heat.¹¹ In particular, nanoparticles made of plasmonic materials (e.g., Au, Ag, and Cu) interact very strongly with light through localized surface plasmon resonances (LSPRs).^{12,13} These nanoparticles initially convert light energy into high-energy electrons and holes that, over timescales of picoseconds, thermalize into heat. The heat generation occurs as the electrons couple to the phonons in the crystalline structure of the nanoparticle, vibrationally exciting its lattice. Under illumination, plasmonic particles can convert photons into highly confined and intense thermal gradients. These “hotspots” serve as a platform to manipulate thermal landscapes at the nanoscale and enable ultrafast heat management on sub-picosecond timescales.^{14,15}

When heat is confined at dimensions approaching molecular and atomic scales, remarkable phenomena emerge that diverge dramatically from conventional macroscopic behavior.^{16,17} At such small scales, thermal energy no longer diffuses uniformly as described by classical Fourier conduction but instead becomes highly localized, transiently stored, and selectively transported.^{18,19}

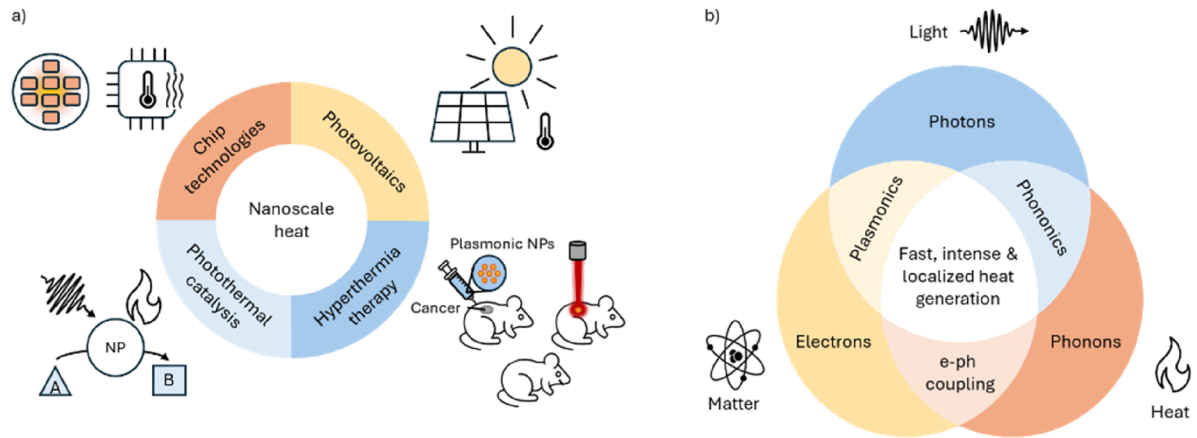


FIG. 1. (a) Examples of industry and research sectors where nanoscale heat management plays a crucial role. (b) Scheme representing how mastering the interplay between photons, electrons, and phonons in a nanoscale system can lead to fast, intense, and localized heat generation.

This nanoscale confinement of thermal energy, both spatially and temporally, opens entirely new pathways to manipulate energy flows at the nanoscale. In this context, the need for accurate predictive modeling becomes critical. Here, the temperature models and the Boltzmann transport equation work as tools to capture the temporal and spatial aspects of plasmonic heat generation.²⁰ This way, the electron and phonon dynamics can then be explored and manipulated to achieve faster and more localized heat generation. In addition, the plasmonic and phononic properties of the systems can be exploited to design layer stacking and substrates to direct thermal flows and selectively heat portions of the sample or device.

Overall, this review provides an overview of strategies that can be employed to generate, control, and measure heat at the nanoscale. Our goal is to chart a path toward extreme nanoscale thermal control and new functionalities in nanophotonic devices. We do this by emphasizing how tailored photothermal material properties, nanostructure design, pulsed optical excitation, and advanced thermal modeling collectively enable the precise control of energy flow at the nanoscale. Therefore, we study the fascinating interplay between photons, electrons, and phonons, i.e., light, matter, and heat transfer at the nanoscale [Fig. 1(b)].

In the section titled Plasmonic resonances and heat generation at the nanoscale, we discuss the plasmonic heat generation mechanism of individual nanoparticles and arrays under continuous wave (CW) and pulsed illumination. In the section titled Non-Fourier heat conduction, we then turn toward available modeling options to accurately describe nanoscale heat conduction in the pulsed and confined regimes. In the section titled Strategies for nanoscale heat generation, we describe material science and nanophotonic strategies to increase the spatial heat confinement. The section titled Nanothermometry outlines different nanothermometry technologies available that are capable of resolving temperature profiles with nm precision and at fs-ps time resolution. Finally, the section titled Conclusion and outlook summarizes and concludes this review.

PLASMONIC RESONANCES AND HEAT GENERATION AT THE NANOSCALE

Plasmonic resonances

When light interacts with matter, it can either be absorbed or scattered. The extent of these processes is typically quantified in terms of optical cross sections,

$$\sigma_{\text{sca}} = \frac{k^4}{6\pi} |\alpha|^2, \quad (1)$$

$$\sigma_{\text{ext}} = k \operatorname{Im}(\alpha), \quad (2)$$

$$\sigma_{\text{abs}} = \sigma_{\text{ext}} - \sigma_{\text{sca}}, \quad (3)$$

where σ_{sca} , σ_{ext} , and σ_{abs} are the scattering, extinction, and absorption cross sections, respectively (m^2); k is the wave number ($k = \frac{2\pi}{\lambda}$, units: m^{-1}); and α is the particle polarizability (m^3). Among photothermal materials, plasmonic nanoparticles (NPs) are particularly appealing nanoscale heat sources thanks to their exceptionally high absorption cross sections at the resonance frequency. This allows them to absorb a large fraction of incident light within a very small volume, which leads to large temperature gradients, i.e., large temperature differences within short distances. In addition, they allow both spatial and temporal confinement of temperature when excited at their resonance frequency.¹²

For a spherical NP, the polarizability is a measure of how strongly the particle is polarized under the external driving electric field. Under the quasi-static approximation, i.e., for nanoparticles whose size is much smaller than the light wavelength, the polarizability of a sphere simplifies to

$$\alpha = 3V \frac{\tilde{\epsilon} - \epsilon_{\text{medium}}}{\tilde{\epsilon} + 2\epsilon_{\text{medium}}}, \quad (4)$$

where V (m^3) is the volume of the particle, $\tilde{\epsilon} = \epsilon_1 + i\epsilon_2$ is the complex

permittivity of the metal (no unit), and ϵ_{medium} is the permittivity of the surrounding medium. In this expression, the polarizability maximizes when the magnitude (modulus) of the denominator minimizes, which occurs for $\epsilon_1 = -2\epsilon_{\text{medium}}$, also known as the Fröhlich or resonance condition. In turn, this maximizes the scattering and absorption cross sections. This phenomenon, which manifests itself as a strong optical extinction at wavelengths characteristic of the considered metal, is called localized surface plasmon resonance (LSPR). Plasmonic nanoparticles can have absorption cross sections larger than their geometric cross section (πr^2), making them ideally suited to concentrate light absorption in extremely small volumes and generate steep nanoscale temperature gradients.

The process by which light absorption is converted into heat in plasmonic nanoparticles is schematically depicted in Fig. 2(a). When a metal nanoparticle is irradiated at the LSPR wavelength, light absorption gives rise to highly energetic electrons and holes. Indeed, following absorption, the LSPR dephases via Landau damping, which couples the collective plasmonic oscillation to single-electron

excitations. Depending on the spectral overlap between the LSPR resonance and the electronic band structure of the material, this dephasing can drive either intraband or interband transitions (or both), giving rise to higher-energy carriers whose energies are determined by the precise plasmon-band overlap. Briefly, in an interband absorption, photons promote electrons from lower-lying energy bands into the conduction band, creating hot holes deep below the Fermi level and relatively “less-energetic” electrons above it. In intraband absorption, the excitation occurs within the conduction band, yielding “more-energetic” electrons.^{21,22} Overall, these highly energetic charge carriers have extremely short lifetimes (<100 fs) and quickly redistribute their energy via electron–electron (e–e) scattering, becoming less-energetic “hot” carriers. These thermalized carriers then undergo electron–phonon (e–ph) coupling and transfer energy to the phonons in the lattice of the plasmonic NP. Finally, the thermal relaxation of the lattice allows for heat to be transferred to the surroundings, generating a temperature gradient surrounding the nanoparticle.^{23,24}

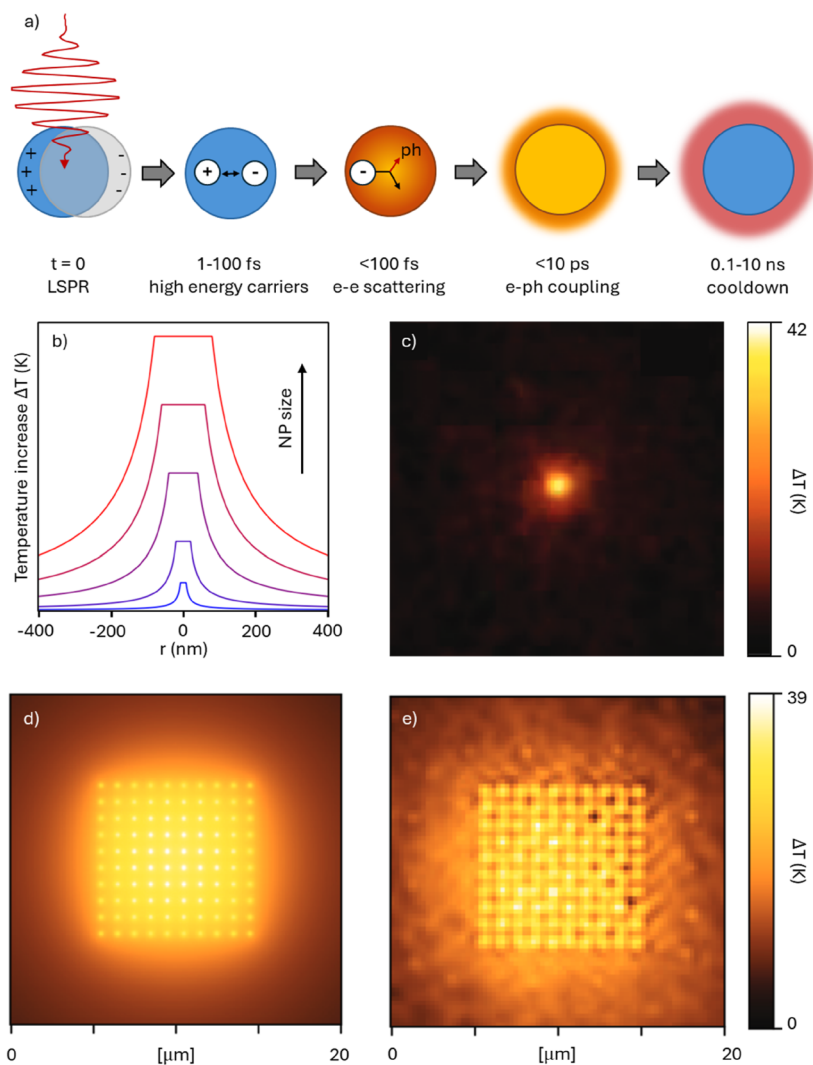


FIG. 2. (a) Steps of plasmonic heat generation and timescales of the involved mechanisms. (b) Temperature gradients surrounding nanoparticles with different radii using Eq. (15). (c) Thermal landscape of a single Au disk (diameter $d = 300$ nm and thickness $h = 40$ nm) illuminated by a laser beam at wavelength $\lambda = 808$ nm and irradiance $I = 12.8$ mW/ μm^2 measured by thermal imaging using quadriwave shearing interferometry (TIQSI). Reprinted with permission from Baffou *et al.*,²⁵ ACS Nano 6(3), 2452 (2012). Copyright 2012 American Chemical Society. (d) Numerical simulation and (e) TIQSI thermal imaging of a pattern of 10×10 gold nanodisks with 200 nm diameter, 40 nm height, and 1 μm pitch. Illumination wavelength was $\lambda = 750$ nm, irradiance not specified. Reprinted with permission from Baffou *et al.*,¹⁵ ACS Nano 7(8), 6478–6488 (2013). Copyright 2013 American Chemical Society.

The timescales at which these mechanisms occur are important parameters to consider when designing heat generation at the nanoscale and have been well studied, especially under pulsed conditions using time-resolved spectroscopy.²⁶ Hot carrier generation upon irradiation takes between 1 and 100 fs in noble metals, which is quickly followed by e-e scattering (<100 fs) and e-ph coupling (≤ 10 ps). Here, it is important to realize that these electron dynamics can be strongly illumination specific: for low-intensity CW illumination, hot electrons can live up to 1 ns, with a strongly suppressed electron-phonon coupling rate.²⁷ Finally, the cooldown of the nanoparticle takes a longer time, typically ranging between 0.1 and 10 ns according to the effective thermal conductivity of the surrounding medium.²⁶

Continuous wave illumination

Under continuous wave illumination, the temperature increase in a nanoparticle results from the balance between the rate of photothermal heating (via light absorption) and its rate of cooling (via heat conduction, convection, and thermal radiation to the surrounding environment). In solids, heat transport is generally governed by Fourier's law, which neglects convection and radiative losses and defines the conductive heat flux density q (W/m^2) as

$$q(r, t) = -\kappa \nabla T(r, t), \quad (5)$$

where κ is the thermal conductivity ($W/m K$) and ∇T is the spatial temperature gradient (K/m). Fourier's law by itself describes how heat flows in response to a temperature gradient, but it does not describe how temperature evolves in time. To do so, it must be combined with the principle of energy conservation, which ensures that the change in stored energy over time is equal to the sum of heat flux to the surroundings and the internal heat generation. This conservation principle for a stationary solid is expressed as^{28–30}

$$\frac{\partial(\rho C_p T)}{\partial t} = -\nabla \cdot (q(r, t)) + Q, \quad (6)$$

where ρ is the density (kg/m^3), C_p is the specific heat capacity ($J/kg K$), and Q is the volumetric heat generation rate (W/m^3). Substituting Fourier's law [Eq. (5)] into Eq. (6) gives the classical heat diffusion equation, which describes how temperature evolves spatially and temporally in response to internal heating and thermal conductivity,³¹

$$\frac{\partial(\rho C T)}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q. \quad (7)$$

To apply this thermal model to an isolated spherical nanoparticle under CW illumination, we consider the steady-state limit where the temperature no longer changes in time. In this case, the left-hand side of Eq. (7) vanishes, and the equation reduces to a balance between internal heat generation and heat dissipation via conduction,

$$\nabla \cdot (\kappa \nabla T) = -Q. \quad (8)$$

For an isolated spherical nanoparticle embedded in an infinite, homogeneous medium, the heat generation rate Q is uniform inside

the particle (for $r \leq R$) and zero outside. The temperature profile outside the nanoparticle (for $r \geq R$) satisfies Laplace's equation,

$$\nabla^2 T = 0. \quad (9)$$

Assuming spherical symmetry, this simplifies to

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0. \quad (10)$$

Since the derivative of $r^2 \frac{dT}{dr}$ is 0, its integral is a constant (C_1), leading to

$$\frac{dT}{dr} = \frac{C_1}{r^2}. \quad (11)$$

Integrating a second time gives the general solution,

$$T(r) = \frac{C_1}{r} + C_2. \quad (12)$$

Applying the boundary condition that the temperature approaches ambient far away from the particle [$T(r) \rightarrow T_{amb}$ for $r \rightarrow \infty$], the integration constant C_2 is recognized to be T_{amb} . The constant C_1 is determined by energy conservation as follows: the total power conducted through the spherical shell of the nanoparticle (P_{out}) with area A (m^2) is given by Fourier's law,

$$P_{out} = -\kappa A \left. \frac{dT}{dr} \right|_{r=R} = -\kappa \cdot 4\pi R^2 \cdot \left(\frac{C_1}{R^2} \right) = -4\pi \kappa C_1. \quad (13)$$

Next, the total absorbed power (P_{abs} , W) can be found from the product of the absorption cross section and the incident light intensity I (W/m^2), that is, $P_{abs} = \sigma_{abs} I$, which is assumed to be spatially uniform. This absorbed power must equal the total dissipated power ($P_{out} = -P_{abs}$), so that

$$4\pi \kappa C_1 = \sigma_{abs} I \Rightarrow C_1 = \frac{\sigma_{abs} I}{4\pi \kappa}. \quad (14)$$

The final substitution gives the spatial temperature profile around the nanoparticle,³²

$$T(r) = \frac{\sigma_{abs} I}{4\pi \kappa r} + T_{amb}, \quad r \geq R. \quad (15)$$

From Eq. (15), consequently, the spatial temperature gradient around an irradiated plasmonic nanoparticle decays as $1/r$ [Fig. 2(b)]. On the other hand, the temperature inside the nanoparticle can be assumed to be constant when the material's thermal conductivity is much larger than that of the surroundings and/or in the presence of a high interfacial heat transfer resistance.^{28,33} It can, therefore, be assumed that the temperature inside the NP, i.e., for $r \leq R$, will be constant and equal to

$$T_{NP} = T(R) = \frac{\sigma_{abs} I}{4\pi \kappa R} + T_{amb}. \quad (16)$$

To extend this model to non-spherical nanoparticles, a correction factor (β) must be included in the denominator,³⁴

$$T_{NP} = \frac{\sigma_{abs} I}{4\pi \kappa R_{eff} \beta} + T_{amb}, \quad (17)$$

where R_{eff} is the radius of a corresponding sphere with the NP's volume and $\beta \neq 1$ for non-spherical NPs.³⁵ The validity of Eqs. (15)–(17) for plasmonic nanoparticles on substrates has been experimentally proven by comparing their theoretical values with experimentally determined temperature profiles using thermometric imaging. For example, Fig. 2(c) clearly demonstrates the radial temperature profile around a single gold nanodisk under 808 nm excitation ($I = 12.8 \text{ mW}/\mu\text{m}^2$), which was accurately predicted by Eq. (17).²⁵

Collective heating in nanoparticle arrays

However, when considering the illumination of an array of nanoparticles, T_{NP} additionally depends on the collective heating by the surrounding particles, which becomes the dominant contribution for closely spaced arrays.¹⁵ Equation (16) is, therefore, modified to include the temperature increase as a result of collective heating ($\Delta T_{\text{collective}}$).³²

$$T_{\text{NP}} = \frac{\sigma_{\text{abs}} I}{4\pi\kappa R} + T_{\text{amb}} + \Delta T_{\text{collective}}. \quad (18)$$

In an infinite assembly, $\Delta T_{\text{collective}}$ is the sum of the heat delivered by the $N - 1$ other NPs surrounding the central nanoparticle j , as expressed by¹⁵

$$\Delta T_{\text{collective}} = \sum_{k=1, k \neq j}^N \frac{q_k}{4\pi\kappa} \frac{1}{|r_j - r_k|}, \quad (19)$$

where q_k is the heat power (W) delivered by nanoparticle k located at r_k . From Eq. (19), it is possible to obtain an approximation for the collective heating in the limit of large particle numbers. The exact expression depends on the nature of the array and the illumination conditions.^{15,36} For example, for a uniformly illuminated finite-sized square array, $\Delta T_{\text{collective}}$ is expressed as

$$T_{\text{center}}^{\text{collective}} \approx \frac{\sigma_{\text{abs}} I}{\kappa} \frac{\ln(1 + \sqrt{2})}{\pi} \frac{S}{p^2}, \quad (20)$$

where S is the side length of the square array (m) and p is the interparticle distance (pitch, in m). To prove the validity and universality of these approximations, thermal imaging and numerical simulations were performed on a broad variety of finite and infinite gold nanoparticle arrays, such as an array of 10×10 gold nanodisks [Figs. 2(d) and 2(e)]. Indeed, in all cases, the temperature of the central nanoparticles could be accurately predicted using Eq. (18).¹⁵ These equations, therefore, provide a straightforward yet versatile framework for quantitatively predicting and engineering both local and collective temperature rises in nanoparticle assemblies.

Nonetheless, this model relies on the relatively simple and linear Fourier-based heat diffusion model and assumes that each NP can absorb light independently and equally throughout the assembly. Hence, the applicability of such a model becomes challenging for actual samples, where these conditions are not met. Indeed, this model becomes rather complex and begins to break down when the array becomes, for example, three dimensional. For such 3D arrays, the depth-dependent attenuation of the excitation beam cannot be neglected anymore, and heat conduction may sum over many layers of NPs.³⁷ This also happens when the shape of the nanoparticles is

not perfectly spherical and when the interparticle spacing becomes so tight that optical or thermal interactions are strong and need to be accounted for. It was shown that for an individual NP, the photothermal temperature increase can be described using an “effective radius,” even if the shape is not strictly spherical.³⁵ This approximation models the relation between the power absorbed by the NP and the local temperature elevation. However, for very tight clusters or irregular 3D shapes of particles, or for complex morphologies, this effective radius approach may lead to inaccuracies.

Pulsed illumination

The previous equations only apply to steady-state CW illumination. Under pulsed irradiation, the heating and cooling dynamics of metal nanoparticles change dramatically. Typical laser pulses can have durations as short as hundreds of femtoseconds and repetition rates in the 10^7 Hz range. These timescales strongly overlap with the heating and cooling dynamics of plasmonic nanoparticles [Fig. 2(a)], making their temperature evolution highly dependent on the laser characteristics. However, the Fourier heat diffusion is valid once the e-ph coupling and phonon-phonon scattering have equilibrated (>100 ps), and the thermal properties of the medium behave diffusively. When operating at ps or fs timescales, the electron gas in each NP can undergo rapid, non-Fourier (ballistic or nonlocal) heat flow before the usual continuum diffusion has time to set in. Since the pulses arrive faster than the cooling rate of the NP, the resulting temperature can accumulate and lead to a quasi-steady “average” heating. The peak “instantaneous” temperatures achieved with each pulse exceed significantly the predictions of a purely steady, continuous-wave model and can dramatically change the local thermal environment.

An expression for the temperature profile generated by fs-pulsed optical heating of gold nanoparticles was derived,³⁸ where the peak temperature distribution is the point-by-point (i.e., at each distance from the surface of the NP) maximum of the transient temperature rise outside the NP. This temperature envelope is defined at a delay with respect to the light pulse as it follows the transient rise and fall of the temperature at a fixed distance rather than measuring the temperature right when the light pulse ends. It reads

$$T_{\text{peak}}(r) = \frac{1}{3\sqrt{3}} \frac{E}{C_p \rho} \frac{1}{r^3}, \quad (21)$$

where E is the absorbed energy (J). Here, the peak temperature distribution scales with $1/r^3$, which represents a significant increase in confinement under fs-pulsed illumination compared to CW, which scales as $1/r$ [Eq. (15)]. However, this spatial confinement is transient. The time-averaged temperature distribution of a single NP under pulsed illumination still scales as $1/r$.¹⁵ Therefore, it is important to distinguish between the instantaneous peak and the time-averaged distribution and understand when the latter becomes valid.

This transition is governed by the thermal diffusivity of the surrounding medium (α). The characteristic timescale τ at which thermal energy diffuses from the NP to a point at distance r can be estimated by $\tau \sim \frac{r^2}{4\alpha}$, where $\alpha = \frac{\kappa}{\rho C_p}$. Before this time, the temperature at point r remains largely unaffected by the heat source. After τ , the thermal front arrives and the temperature begins to rise while the

spatial temperature profile starts to evolve from a $1/r^3$ to a $1/r$ distribution. For example, for a pulse-heated Au nanosphere in water, the diffusion time at 1 μm distance is 1.7 μs . For such typical systems (nanoparticles in water and on glass), the steady-state solution ($1/r$) is obtained for $t > 1$ ms, which is highly dependent on the sample surroundings.^{39,40} In general, this relationship dictates that the spatiotemporal confinement under pulsed illumination is retained longer for media with low thermal diffusivity. It also highlights that local temperature measurements under CW and pulsed conditions will lead to identical results, unless fast time-resolution is available.

NON-FOURIER HEAT CONDUCTION

The breakdown of the Fourier model

The Fourier heat equation usually describes plasmonic heating and heat conduction very well, but the discussion above highlights

some of its limits. Indeed, the Fourier heat transport formalism breaks down at very small dimensions and at short time scales. In this section, we outline the various other models that are available to reach a more accurate description of plasmonic systems, both at the nanoscale and under pulsed illumination.

The Fourier model relies primarily on three main assumptions [Fig. 3(a)].^{28,41} (i) Every infinitesimal volume element of the material is assumed to be in local thermodynamic equilibrium, meaning that the macroscopic temperature can fully describe the state of the material at every point. This also implies that the material is treated as a continuous medium where temperature changes occur gradually, ignoring the atomic or molecular structures. (ii) Fourier's law also assumes a linear relation between the driving force (the temperature gradient) and the flux (heat conduction) and the instantaneous response. Indeed, the parabolic nature of the Fourier heat equation implies that for any $t > 0$, the temperature is nonzero at all possible

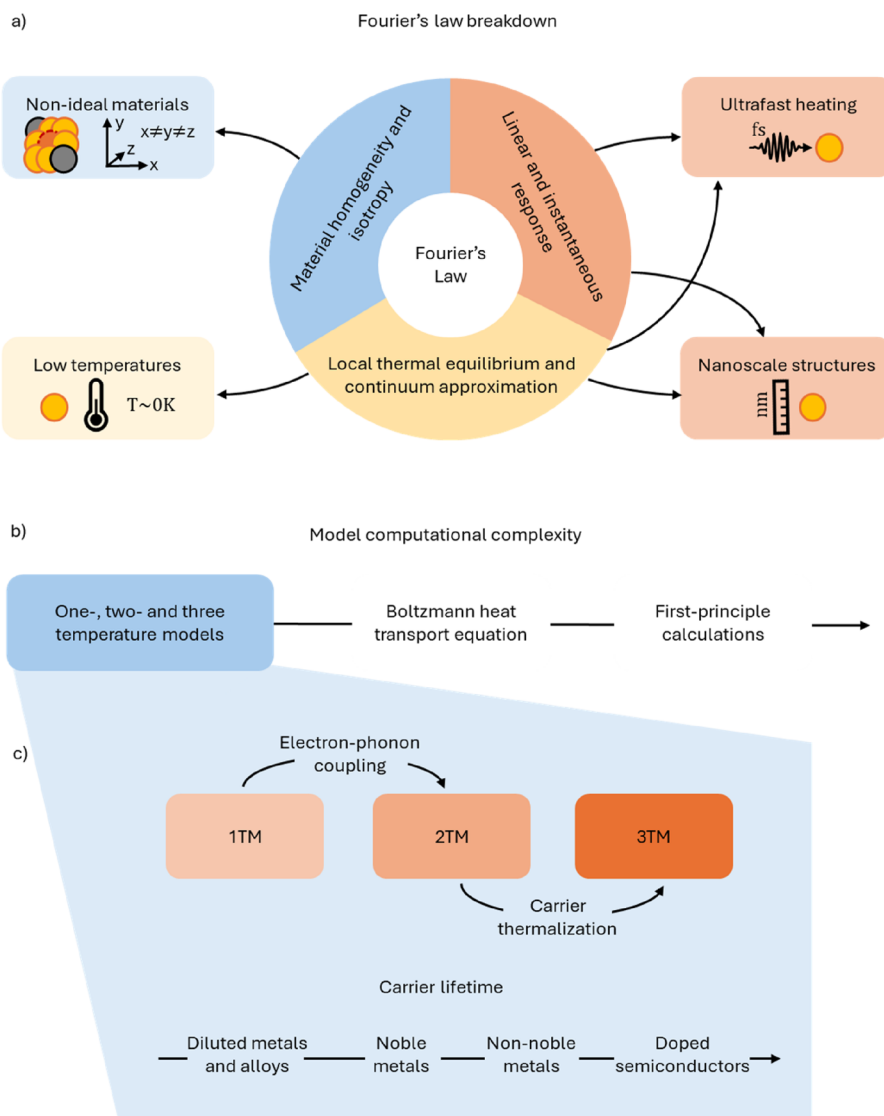


FIG. 3. (a) Key assumptions of Fourier's law and conditions under which they no longer hold. (b) Main models employed to study heat conduction at the nanoscale, ordered on a scale from less to more complex. (c) The temperature models commonly employed to study plasmonic processes, highlighting the aspect of the process they account for. On the bottom, different classes of materials are ordered based on their carrier lifetimes and which model can represent them well.

points, implying that the heat instantaneously diffuses over an infinite distance. (iii) Finally, material homogeneity and isotropy are assumed, which means that the thermal conductivity is a constant or a well-defined function that does not vary abruptly in all directions within the material.

These assumptions lead to significant limitations in the applicability of the Fourier model to cases in which thermal gradients vary abruptly over length scales commensurate with molecular dimensions or when heating and cooling rates exceed typical heat equilibration times.²⁹ For heat transport to remain firmly within the diffusive regime, the characteristic size L of the system must be much larger than the phonon mean free path Λ ($L \gg \Lambda$), ensuring that carriers undergo multiple scattering events to transfer heat. For $L \sim \Lambda$, carriers travel ballistically, i.e., with little to no scattering events occurring during transport. Ballistic transport contributions increase at the nanoscale, so that the Fourier model often fails to capture nanoscale heating and cooling processes. Recently, the ballistic and diffusive mean free paths of electrons in gold were measured.⁴² Ballistic transport of hot electrons was found to extend up to ~ 200 nm immediately after excitation (i.e., before electron–phonon equilibration). Once the electrons thermalize among themselves, their diffusive mean free path settles to roughly ~ 45 nm. The Fourier model is also only valid up to a certain temperature limit. For example, if the temperature were to approach $T \sim 0$ K, the number of phonons would decrease significantly, leading again to ballistic instead of diffusive transport.

In addition, the system under study must have enough time to establish a local thermal equilibrium to be properly described by Fourier's heat transport equations. If the timescale of the heating process becomes comparable to the relaxation time of the carriers ($t \gg \tau$), heat transfer starts behaving as a wave-like phenomenon, making the Fourier model unreliable. Since the plasmonic nanoparticles under pulsed illumination feature both steep nanoscale temperature gradients and ultrafast heating and cooling dynamics, predicting their temperature evolution requires a different, non-Fourier conceptual approach.⁴³

Two and three temperature models

As previously discussed, (ultra)fast pulsed illumination at fs–ns duration can be used to achieve fast, intense, and localized temperature generation. The electron–electron scattering (0.1–1 ps) in noble metals is faster than electron–phonon coupling (1–10 ps). Hence, initially, only the electrons are excited and experience a temperature increase. The evolution of this system can be described using two-temperature models (2TMs),^{44,45} which distinguish between the electron temperature (T_e) and the lattice temperature (T_l).⁴⁶ There are many forms of the 2TM, which have been critically reviewed before,³¹ the most straightforward of them being the parabolic 2TM. Here, two Fourier heat equations linked by an e–ph coupling constant are employed to represent the separate electron and lattice temperatures (T_e and T_l , respectively). The model then reads⁴⁷

$$C_e \frac{\partial T_e}{\partial t} = \nabla \cdot (\kappa_e \nabla T_e) - G(T_e - T_l) + Q, \quad (22)$$

$$C_l \frac{\partial T_l}{\partial t} = \nabla \cdot (\kappa_l \nabla T_l) + G(T_e - T_l), \quad (23)$$

where C_e and C_l are the volumetric electron and lattice heat capacities (J/K m^3), κ_e and κ_l are the electron and phonon thermal conductivities (W/m K), respectively, G is the e–ph coupling constant ($\text{W/m}^3 \text{ K}$), and Q is the absorbed optical power density (W/m^3).

The 2TM offers a relatively simple approach to tackle the dynamics of plasmonic systems, especially compared to theoretical first-principles calculations such as the ones relying on time-dependent density functional theory (TD-DFT) and the time-dependent Boltzmann equation (BE) [Fig. 3(b)].^{20,48} In addition, when compared to the classical one temperature model (1TM, i.e., the Fourier model), the 2TM can capture the process of electron–phonon scattering very well. This feature makes it a particularly interesting tool to model plasmonic processes that involve very fast e–ph dynamics, such as in metal alloys and composites,^{49–51} noble metals,^{52,53} and non-noble metals [Fig. 3(c)].^{54,55} However, it is fundamentally limited by its assumption of instantaneous thermalization of the electrons, which does not reflect the actual dynamics of carriers in plasmonic systems. Indeed, this model does not take into account the initial high energy, non-thermal distribution of the electrons generated upon illumination. Therefore, dynamics at fs timescales cannot be accurately modeled, and it must be assumed that the thermalized distribution after 1 ps is representative.

To overcome this limitation, while keeping a balance between computational complexity and accuracy, the 2TM can be extended with a third energetic bath. This third bath represents the excess energy density stored in the population of out-of-equilibrium carriers (N , in J/m^3) that couple to both the electrons and the phonons.^{48,56} This approach is commonly referred to as the three-temperature model (3TM), even though, strictly speaking, there is no third temperature introduced. Whereas the most basic 3TM models disregard the spatial dependence of N , T_e , and T_l , because they assume a homogeneous absorption across the nanostructure, this does not occur in the case of large nanostructures,^{52,57} complex shapes,^{58,59} and particular excitation conditions.⁶⁰ For such situations, an inhomogeneous 3TM (I3TM) proved to be a better representation of the systems considered.^{61,62} In general, when excitation results in strongly localized light absorption and hot carrier excitation, and the diffusion length of these hot carriers within the thermalization time is shorter than the dimensions of the structure, the inhomogeneous 3TM (I3TM) should be used. The I3TM takes the form of

$$\frac{\partial N}{\partial t} = Q - aN - bN, \quad (24)$$

$$C_e \frac{\partial T_e}{\partial t} = aN - \nabla \cdot (\kappa_e \nabla T_e) - G(T_e - T_l), \quad (25)$$

$$C_l \frac{\partial T_l}{\partial t} = bN + \nabla \cdot (\kappa_l \nabla T_l) + G(T_e - T_l), \quad (26)$$

where a is the electron gas heating rate governing the energy dynamics between non-thermalized and thermalized electrons and b is the decay rate from out-of-equilibrium non-thermal electrons to phonons, both in s^{-1} . The spatial dependence for N , T_e , and T_l was omitted for clarity. For complex and non-symmetric systems, this system of differential equations should be numerically solved.^{61,63}

Although the I3TM captures the spatiotemporal evolution of energy in plasmonic systems, it relies on an instantaneous-dephasing approximation and, therefore, overlooks the energy-dependent

generation and spectral distribution of hot carriers produced during plasmon decay. However, photon absorption and LSPR excitation are distinct and temporally separated processes. Thus, by assuming instantaneous dephasing, the I3TM, therefore, misses this energy-selective hot-carrier generation pathway.⁶⁴ In actuality, different hot carrier generation mechanisms take place when the photon energy exceeds the interband threshold.⁶⁵ Above this threshold, the Landau-damped direct transitions inject highly energetic electrons according to a distribution that is sensitive to the electronic band structure. In contrast, below the threshold, phonon- and surface-assisted processes generate a nearly flat distribution of hot carriers.⁶⁵ Nevertheless, the I3TM effectively captures the concept of carrier lifetimes and mean free path by introducing the dynamics of both non-thermalized and thermalized carriers. This feature makes the three temperature models particularly applicable to plasmonic processes that involve materials with distinct and long carrier lifetimes [Fig. 3(c)], such as doped semiconductor nanocrystals.^{66,67} All these considerations become particularly important when selecting the material and shape of the plasmonic system. The interplay between hot carrier generation, their lifetimes and mean free paths, electron-phonon coupling, and heat transfer at the nanoscale deeply affects the heat generation intensity, confinement, and effectiveness.

Overall, the temperature models (2TM/3TM) extend the Fourier model by introducing separate energy carriers and their finite-rate interactions, but they retain the basic assumptions and additionally introduce parameter uncertainties of their own. These limitations are inherited, as the temperature models maintain the mathematical form of the 1TM, the Fourier heat transport equation. Arguably, they are not strictly non-Fourier models but rather extensions of the 1TM. Nevertheless, these models represent the carrier dynamics over time quite effectively, making for valuable and widely applied tools to model ultrafast processes.

The Boltzmann transport equation and extended models

Although temperature models capture ultrafast plasmonic carrier dynamics by building on Fourier's one-temperature framework, they inherit its unphysical limitations, i.e., the neglect of the different nanoscale transport modes. As device dimensions shrink, ballistic effects increasingly dominate over diffusion, yet these models still treat heat flow as purely diffusive. An intermediate transport regime that is neither fully ballistic nor diffusive is obtained when the characteristic length of the nanostructure approaches the mean free path of the carriers, and it is called the ballistic-diffusive regime. In this regime, it is a common approach to model nanoscale heat transport through the Boltzmann transport equation (BTE), which can be considered an actual non-Fourier model.^{43,68–72} Since both electrons and phonons contribute to nanoscale heat transport,⁷³ the BTE equations for both carriers must be computed. The electrons and phonons distributions evolve over time because of three main effects: drift, external forces (for electrons), and scattering (collisions with other particles or boundaries). Meanwhile, sources can inject new carriers,

$$\frac{\partial f_k^e}{\partial t} + v_k \cdot \nabla f_k^e + F \cdot \nabla_k f_k^e = \left. \frac{\partial f_k^e}{\partial t} \right|_{\text{scatt}} + s^e, \quad (27)$$

$$\frac{\partial n_{q,p}^p}{\partial t} + v_{q,p} \cdot \nabla n_{q,p}^p = \left. \frac{\partial n_{q,p}^p}{\partial t} \right|_{\text{scatt}} + s^p. \quad (28)$$

Here, $\frac{\partial f_k^e}{\partial t}$ represents the rate of change of the occupation probability of electrons in state k . The second term ($v_k \cdot \nabla f_k^e$) represents the drift of electrons in real space due to spatial gradients in the distribution, where v_k is the group velocity of electrons (m/s) and ∇f_k^e is the spatial gradient of the electron distribution (m^{-1}). The third term ($F \cdot \nabla_k f_k^e$) represents the redistribution of electrons in momentum space due to an external force F (N), such as from an external electric field. (e.g., from an electric field). This term can be coupled to the electric field during plasmonic excitation (obtained from Maxwell's equations), which perturbs the electron distribution. On the right hand side of Eq. (27), $\partial f_k^e / \partial t|_{\text{scatt}}$ represents the rate change introduced by scattering, which includes electron-electron, electron-phonon, and electron-surface scattering processes. Finally, s^e represents the rate of electron excitation due to absorbed light energy.

To describe the phonon transport [Eq. (28)], $\frac{\partial n_{q,p}^p}{\partial t}$ is the rate of change in the phonon occupation number for wavevector q and branch p . The second term ($v_{q,p} \cdot \nabla n_{q,p}^p$) represents the diffusion of phonons, where $v_{q,p}$ is the group velocity of phonons (m/s) and $\nabla n_{q,p}^p$ is the spatial gradient of the phonon distribution (m^{-1}). Furthermore, $\partial n_{q,p}^p / \partial t|_{\text{scatt}}$ is the phonon rate change caused by scattering. Finally, s^p is the rate at which hot electrons transfer energy to the lattice via electron-phonon scattering.

Together, Eqs. (27) and (28) provide a comprehensive semiclassical framework for tracking how the distribution of electrons and phonons evolves in space and time. In contrast to the many-temperature models, the more complex BTE acknowledges the finite speed of carrier movement and includes both diffusive and ballistic heat transport modes that play a large role at the nanoscale. This method is, therefore, inherently more accurate in the description of nanoscale heat transport. While it can, in principle, describe non-equilibrium carrier and phonon dynamics, its direct application to plasmonic systems is still limited, as it does not fully capture the quantum nature of plasmon decay or the highly non-thermal distribution of hot carriers immediately following excitation. Further challenges lie in dealing with a strong spatial confinement and obtaining reliable scattering rates and band structures, which are often unavailable for nanostructured and heterogeneous systems. For this reason, plasmonic heat transfer studies commonly rely on models with reduced complexity. In particular, when electrons and the lattice can be considered as two coupled thermal reservoirs with well-defined temperatures, the BTE formalism simplifies to the 2TM.⁷³

Finally, several extended heat conduction models have been developed as approximations to the phonon Boltzmann transport equation, which are particularly relevant in systems exhibiting ballistic or quasi-ballistic transport.⁴³ To address such non-diffusive effects, models such as the Cattaneo-Vernotte and Guyer-Krumhansl formulations introduce finite thermal propagation speeds, spatial nonlocality, and temporal memory, while remaining consistent with kinetic theory and the second law of thermodynamics.^{29,30,43,74} Solving these generalized models remains complex due to the need for implementing realistic initial and

boundary conditions, obtaining geometry-specific scattering rates, accurately describing phonon properties, and accounting for different non-linearities. This makes it difficult to generalize or standardize them. It is, therefore, fundamental to correctly interpret the restrictions of a specific model when applying it to a particular scenario.³⁰ In hybrid plasmonic systems, these generalized models may offer a more appropriate framework to describe heat dissipation in the surrounding medium once thermal energy has been transferred from the excited electron system to the lattice, going beyond the limits of classical diffusion.

STRATEGIES FOR NANOSCALE HEAT GENERATION

Having established the principles of plasmonic heat generation and spatiotemporal heat transfer modeling, we now turn to state-of-the-art strategies for nanoscale and ultrafast thermal control. For many thermoplasmonic applications it is crucial to balance four interlocking factors: (i) the light-to-heat conversion efficiency at a target wavelength or for a spectral bandwidth, i.e., how much of the incident (pulsed) energy converts to localized heat; (ii) the rate of heat delivery, meaning adequate e-ph coupling and thermal

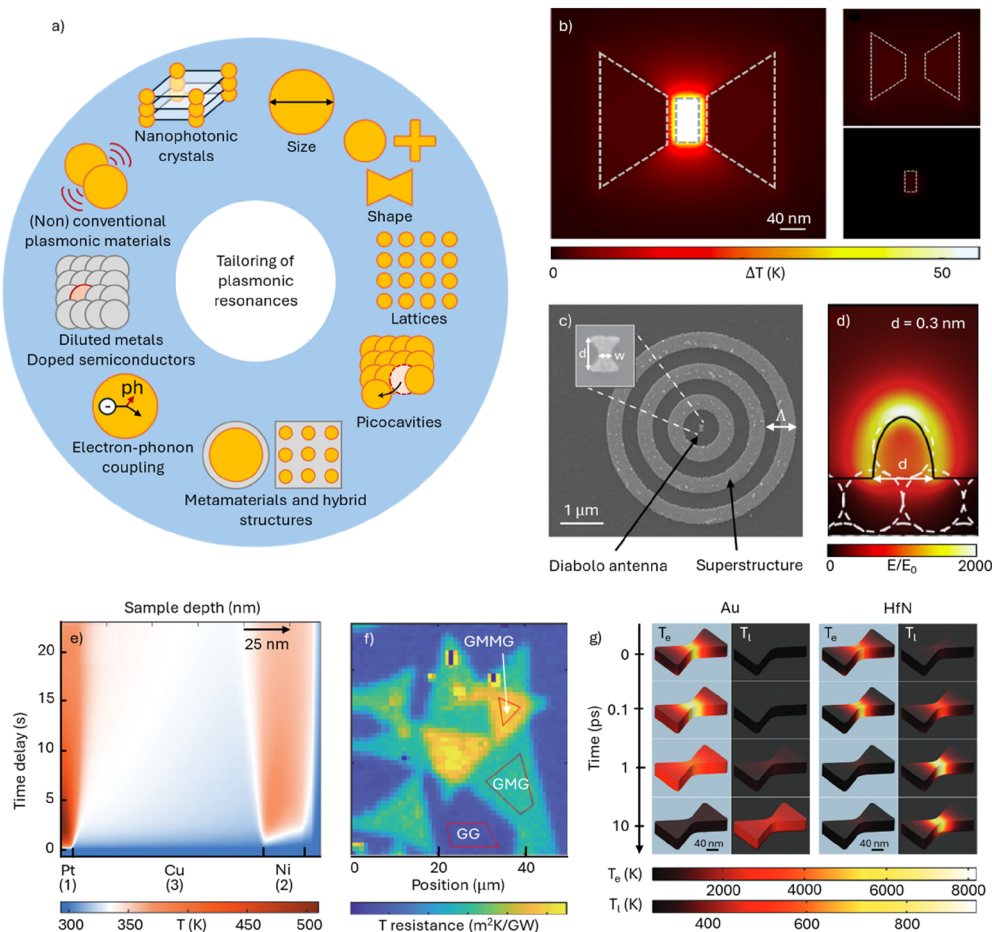


FIG. 4. (a) Strategies to tailor heat generation and conduction at the nanoscale. (b) Simulated temperature increase in a tungsten nanorod including a gold bow-tie antenna (left), just a gold antenna (top right), and just a tungsten nanorod (bottom right). Reprinted with permission from Dongare *et al.*, ACS Nano **15**(5), 8761–8769 (2021).⁷⁷ Copyright 2021 American Chemical Society. (c) SEM image of a plasmonic superstructure made up of a diabolo antenna (150 nm long and wide, 50 nm thick, and 50 nm wide at waist) surrounded by three circular gratings ($\Lambda = 720$ nm). Reprinted with permission from Coppens *et al.*, Nano Lett. **13**(3), 1023–1028 (2013).⁷⁸ Copyright 2013 American Chemical Society. (d) Simulated electric field enhancement of a picocavity, modeled as an ellipsoidal protrusion from a 80 nm Au sphere. Reprinted with permission from Griffiths *et al.*, ACS Photonics **8**(10), 2868–2875 (2021).⁷⁹ Copyright 2021 American Chemical Society. (e) Modeled phonon temperature (2TM) of heat transport in a Pt/Cu/Ni thin-layer stack under pulsed illumination. The numbers represent the order of heating of the layers. Reprinted with permission from Pudell *et al.*, Adv. Funct. Mater. **30**(46), 2004555 (2020).⁸⁰ Copyright 2021 Wiley Advanced. (f) Map of the thermal resistance of stacks of monolayer graphene (G) and monolayer MoS₂ (M), measured by time-domain thermoreflectance (TDTR). Reprinted with permission from Sood *et al.*, ACS Nano **15**(12), 19503–19512 (2021).⁸¹ Copyright 2021 American Chemical Society. (g) Evolution of electronic (T_e) and lattice temperature (T_l) during fs-pulsed excitation of a diabolo nanoreactor made of either Au (left) or HfN (right), highlighting the different transient heat-localization capacity. Reprinted with permission from Askes and Garnett, Adv. Mater. **33**(49), 2105192 (2021).⁵⁹ Copyright 2021 Wiley Advanced.

transport; (iii) the degree of sub-wavelength heat localization, where higher confinement can lead to higher temperatures; and (iv) the structural stability, so that the nanostructure is able to withstand demanding thermal, physical, and chemical conditions. For example, in targeted photothermal therapy using gold nanorods, high light-to-heat efficiency allows using low light intensity, sub-wavelength localization can prevent damage to surrounding healthy tissue, and high structural stability ensures that the nanorod's resonance remains aligned with the light source.⁷⁵ For ultrafast magnetic recording, it is particularly important to use sub-picosecond light-to-heat conversion and strong nanoscale confinement to obtain high data recording resolution.⁷⁶

Thus, it is clear that the different factors must be tailored to specific applications through a thoughtful design that engineers the shape, size, material properties, and spatial organization of the nanostructure, as well as the thermal dissipation and directionality of the surroundings [Fig. 4(a)]. In this section, we identify emerging design strategies that push heat generation and confinement to dimensions at, or even below, the scale of individual nanostructures, identifying geometries, excitation schemes, and hybrid architectures that maximize local temperature rise within minimal volumes. We also examine material platforms beyond traditional noble metals, such as gold and silver, focusing on how their unique optical responses and carrier-dynamics enable unconventional heat-transfer and localization phenomena. Finally, we examine how the micro-environment can be tailored for delaying or accelerating thermal dissipation.

Tailoring the plasmonic resonance

A first and essential strategy to tailor the photothermal effect is by matching the position of the plasmonic resonance to the optical spectrum of the light source. The frequency and intensity of the plasmon resonance change with the shape, size, and material of the NP, as they directly affect the polarizability.⁸² The simplest approach to change the spectral position of the plasmonic resonance is to change the size of the NP. In general, increasing the size while keeping the aspect ratio constant will redshift and broaden the resonance, because a larger NP will experience a phase difference in the incident field across its surface (retardation).⁸³ In addition, larger NPs scatter more light compared to smaller NPs, because the absorption cross section scales with the volume, but the scatter cross section scales with the volume squared [Eqs. (1)–(4)]. Consequently, scatter will compete with absorption for larger nanostructures.

To achieve the highest heating efficiency (achieving the highest temperature for a given irradiance), one should aim to maximize the absorption cross section while minimizing the NP volume, which is also given as maximization of the Joule number, which is expressed as $J_0 = \frac{\sigma_{\text{abs}} \lambda_{\text{ref}}}{2\pi V}$, where $\lambda_{\text{ref}} = 1240$ nm to make the number unitless.⁸⁴ It is important to realize that, in contrast to the plausible intuition that a higher imaginary permittivity, generally connected with higher absorption losses, leads to a higher heating efficiency, it is, in fact, inversely correlated. It was shown that at resonance, $J_0 \approx |\epsilon_1^{\frac{3}{2}}| / \epsilon_2$, indicating that a higher ϵ_1 and a lower ϵ_2 will maximize J_0 , because the electric field *inside* the NP scales as $1/\epsilon_2$.⁸⁴ Thus, a metal with lower optical losses (e.g., Ag) will, in general, lead to higher localized temperatures.

The most commonly used plasmonic metals are gold (Au) and silver (Ag), thanks to their large available free carrier concentration and small Ohmic losses, which result in sharp plasmon resonances. Optical losses in these metals originate from both inter- and intra-band transitions, with interband transitions typically significantly contributing above 1.8 eV in Au and 3.2 eV in Ag.⁸⁵ For this reason, their sharpest and most intense resonances are found below these energies. A notable drawback of noble metals is that under the typically high fluences of pulsed illumination, gold and silver nanoparticles can undergo reshaping, fragmentation, and aggregation. Indeed, peak temperatures above their melting point (800 °C for Au) can easily be achieved in this illumination regime.^{86,87}

The material can also be specifically designed to optimize photothermal effects. One strategy is to alloy noble metals, such as in Ag–Au, Ag–Pd, or Au–Cu systems, which enables spectral tuning of the resonance and adjustment of optical losses through changes in the electronic band structure and damping mechanisms.⁸⁸ An interesting method to achieve such alloys is via the galvanic replacement reaction between Ag nanostructures and HAuCl_4 in aqueous solution, leading to the formation of hollow Au–Ag nanostructures with tunable morphology and optical response.⁸⁹ Another approach is to use doped semiconductors, such as indium tin oxide (ITO) or aluminum-doped zinc oxide (AZO), where the free carrier concentration can be tuned to control plasmonic resonance in the near- to mid-infrared range with lower losses.^{90,91} Transition metal oxides and nitrides, and chalcogenide compounds also display plasmonic resonances.^{92,93} These materials are particularly appealing, as they can be manipulated with phase and compositional changes to obtain a tunable plasmonic response, covering nearly the whole electromagnetic spectrum.⁹³ These metallic derivatives, including alloys, intermetallics, and conductive compounds, offer additional degrees of freedom for engineering the dielectric function, enabling fine-tuning of plasmonic resonances for application-specific spectral and thermal performance.

While adjusting nanoparticle size and material can tune the LSPR spectrum, three essential shape-design principles enable even finer control. These principles account for the boundary conditions that govern dipolar plasmon oscillations and the way neighboring plasmons interact and couple with one another.⁸² (i) The dipole resonance redshifts with the increase in shape anisotropy, the number of corners, and the sharpness of the corners. (ii) The intensity of the plasmon resonance depends on the symmetry of the NP: the higher the symmetry, the sharper and more intense the resonance. (iii) If the NP can be polarized in more than one way, it will support a number of resonances equal to the polarization directions. Commonly employed shapes include nanospheres, rods, cubes, stars, spherical shells, matryoshkas, cages, bow-ties, donuts, diabolos, and many others.^{34,94,95}

Nanophotonic strategies for localized absorption

The NPs spacing and arrangement are also crucial because they can lead to collective optical and thermal effects. Periodic arrays of nanoparticles can sustain surface lattice resonances (SLRs) emerging from the in-plane interference of the electromagnetic fields scattered by adjacent particles.⁹⁶ When illuminating the array at the SLR wavelength, these collective modes display much narrower linewidths and stronger absorption responses than individual

localized plasmons. These lattice resonances affect light-to-heat conversion in nanoparticle arrays under both pulsed and CW illumination⁹⁷ and can enhance the absorption cross section while requiring less material per unit area. This makes them an interesting tool to tailor nanoscale heat generation and achieve spectral and spatial control via array design.

Another strategy to enhance the absorption cross section of the plasmonic nanostructure is via auxiliary photonic structures [Figs. 4(b) and 4(c)]. These structures are deliberately engineered dielectric or metallic architectures that surround the plasmonic nanostructures to mold the local electromagnetic environment and, hence, boost photothermal conversion. Such “superstructure design” can, therefore, be employed to enhance the obtained temperature increase in a system. For example, plasmonic heterostructures can be fabricated so that the plasmonic antenna concentrates the local electric field on one or more adjacent particles (made from a different material, such as a catalyst), which by themselves only interact weakly with the light.^{77,98} This strategy can lead to highly confined, sub-wavelength heating, which was shown by fabricating a gold bowtie antenna surrounding a central tungsten nanorod, which was additionally thermally insulated [Fig. 4(b)].⁷⁷ In another example, a superstructure made up of a diabolo-shaped nanostructure surrounded by three circular gratings outperformed the single resonant diabolo structure, causing a temperature increase over 5 times greater than the one generated by the single antenna [Fig. 4(c)].⁷⁸

Another auxiliary photonic strategy is the use of Fano resonances, which occur when a narrow “dark” plasmonic mode (weakly radiative or symmetry-forbidden) interferes with a broad, “bright” mode (strongly radiative), producing a characteristic asymmetric line shape and dramatic field enhancement at the resonance wavelength.^{99,100} This Fano interference concentrates optical energy into the reactor structure and results in both a spectrally narrowed absorption peak and an order-of-magnitude increase in local field intensity. The Fano effect translates directly into more efficient and stronger localized heating.¹⁰¹ For example, a Fano-heater trimer can be composed of two large antennas positioned to surround a small nanorod, where the nanorod resonance interferes with the broad dimer mode.⁹⁸ The Fano resonance funnels all the energy into the central nanorod that, therefore, produces extremely localized heating. However, these resonances are very sensitive to the exact geometric positioning, which, therefore, requires precise, complex, and expensive fabrication methods.

Aside from superstructures, it is possible to trap light and obtain plasmonic effects from picocavities, which are atomic-scale optical cavities that originate from single atom protrusions.¹⁰² These individual atomic features are able to highly localize electromagnetic fields within ultrasmall volumes [$<1 \text{ nm}^3$, Fig. 4(d)], and even though they are highly unstable at room temperature, their formation and disassembly can be monitored at cryogenic temperatures.¹⁰³ Picocavities are typically formed within plasmonic nanogaps because of the strong interaction between intensely localized fields and the metallic surface, which causes the migration of surface atoms to form protruding adatoms.⁷⁹ They can transiently enhance the local field further thanks to the so-called lightning rod effect, where the pronounced geometrical curvature caused by the adatom abruptly changes the electrical potential, leading to a non-resonantly induced strong and confined electric field that

superimposes on the plasmonic effect of the nanogap.¹⁰⁴ One of the most commonly employed configurations to obtain picocavities is the nanoparticle-on-mirror (NPoM), where a plasmonic antenna is combined with a metal-insulator-metal (MIM) waveguide placed between a plasmonic antenna and a mirror.⁷⁹ Metallic dimers, bowtie nanoantennas, and other nanostructures separated by sub-nanometer gaps can spontaneously form picocavities as well.¹⁰³ Overall, the generated picocavities are very effective in enhancing the photothermal effects of plasmonics due to their ability to localize light at the atomic scale, opening the path to atomic-scale chemistry and material science.¹⁰⁵

Overall, any nanophotonic strategy that increases the absorption cross section will equally enhance the collective temperature of an array of such structures [Eq. (18)]. Therefore, auxiliary resonances do not automatically improve the contrast between the nanoparticle and the surroundings unless they strongly enhance the local electric field and reduce the dimensions of the heated volume. Furthermore, in large-area and closely spaced arrays, collective heating effects will outcompete localized gradients.¹⁵ Consequently, collective heating may “wash out” nanoscale gradients, yielding a smooth, homogeneous temperature field, which is an effect that must be acknowledged in the engineering of larger-scale devices.

Exploiting electron-phonon coupling

The timescale for e-ph coupling in noble metals (0.1–10 ps)^{23,26,106} stems from the relatively low electron-phonon coupling constant ($G = 2.8 \times 10^{16} \text{ W/m}^3 \text{ K}$ for Au),¹⁰⁷ which is sensitive to the size of the nanoparticle.¹⁰⁸ When the characteristic size of the nanoparticle approaches and/or falls below the bulk electron mean free path, the electronic density of states and the available phonon modes are altered. In sub-mean-free-path particles, the increased electron-surface scattering and phonon confinement tend to accelerate energy transfer to the lattice, shortening the effective e-ph coupling time. Overall, the relatively slow e-ph coupling, combined with the high thermal conductivity of noble metals ($\kappa > 300 \text{ Wm}^{-1} \text{ K}^{-1}$),⁵⁹ results in a fast redistribution of energy among electrons before they are able to transfer heat to their lattice. Consequently, despite confinement of light absorption in sub-particle optical hot spots, heat generation typically occurs evenly throughout the whole particle.

Employing materials with higher values of G can give surprising results and lead to novel approaches to ultra-fast confined heating. For example, a triple layer heterostructure made from platinum (Pt), copper (Cu), and nickel (Ni) was studied using ultrafast excitation, and the heat-induced strain was probed using x-ray diffraction [Fig. 4(e)].⁸⁰ While the light is mainly absorbed in the top Pt layer, the bottom layer (Ni) heats up quickly after. However, heating of the Cu is greatly delayed. Here, the hot carriers generated in the Pt cross the highly conductive Cu layer almost ballistically. Since the Ni has a faster electron-phonon coupling than Cu, the electron thermalization leads to heat generation in the nickel layer and thereby effectively bypasses heating in the copper. Such an example of “heat transport without heating” highlights how the nanoscale organization of materials can offer—sometimes counter-intuitively—opportunities in controlling photothermal effects and selectively heating materials.

When materials with strong electron–phonon coupling are shaped into nanophotonic resonators, it becomes possible to selectively heat the nanostructures at their absorption hotspots.^{55,59} For example, it was shown that under pulsed illumination, a plasmonic HfN diabolito nanostructure reached $3\times$ higher local transient temperatures than an Au analog [Fig. 4(g)]. Even though both nanostructures absorbed the same amount of energy and had a similar absorption profile, the much higher e–ph coupling of HfN ($G \approx 1 \times 10^{18}$ W/m³ K) prevented the diffusion of hot carriers away from the central absorption hotspot. Furthermore, thanks to the low thermal conductivity of HfN ($\kappa \approx 20$ W/m K), the heat remained trapped at the absorption hotspot for 100s of ps. An additional benefit of using transition metal nitride (TMN) materials is that they are additionally thermally and chemically robust, which allows them to withstand transiently high temperatures under pulsed illumination.^{109,110}

Engineering the micro-environment

Heat dissipation to the surroundings is a key parameter to consider in photothermal engineering, which can be tailored by the proper choice of the surrounding medium and supporting substrate.^{111–115} To maximize steady-state temperatures, the surroundings should ideally have a low effective thermal conductivity with a high interfacial heat transfer resistance, and vice versa if the goal is to dissipate heat or prevent thermal damage. This thermal resistance at the boundary between two materials (also known as Kapitza resistance) stems from phonon mismatch, where differences in vibrational spectra and acoustic impedance hinder the transmission of heat-carrying phonons across the interface. Thermal resistance at a nanoparticle/environment interface does not affect the steady-state temperature gradient outside the particle, but it can lead to an increase in the temperature of the particle itself, effectively resulting in a temperature step at the interface.^{2,116} Control over interfacial resistance can be achieved through chemical functionalization, control of substrate roughness and NP contact, thin dielectric spacer layers, and reduction of phonon mismatch between NP and surroundings.

Furthermore, heat dissipation can also be directionally guided by creating an anisotropic thermal pathway, for example, using van der Waals (vdW) heterostructures.^{81,117} Such vdW heterostructures are made of two-dimensional (2D) materials stacked layer-by-layer and held together by van der Waals forces. Each layer retains the individual material lattice and electronic structure and in-plane thermal conductivity, but the cross-plane thermal conductivity is strongly reduced due to vibrational mismatch and loose interlayer coupling. This is particularly well displayed in Fig. 4(f), which shows how the thermal resistance, as measured by time-domain thermoreflectance, dramatically increases for alternating stacks of graphene (G) and Mo₂S (M) layers. The resistance increases up to 200 m² K/GW for GMGMG, corresponding to a remarkable 0.02 W/m K cross-plane conductivity (i.e., lower than air).⁸¹ Such properties allow heat transport to be very efficient in the in-plane direction of the heterostructure while remaining limited in the out-of-plane direction, hence enabling the tailoring of heat flow at the sub-nanometer scale. To the best of our knowledge, vdW materials have not yet been applied to control plasmonic heating, but it is a promising avenue.

A related but more ordered class of materials is superlattices (SLs), which are composed of periodically alternating nanolayers of different materials (semiconductors, oxides, or 2D materials), typically grown via epitaxy.¹¹⁸ This structural periodicity modifies the phonon dispersion relations, introducing bandgaps or reduced group velocities that suppress thermal transport, particularly in the stacking direction. In high-quality SLs, coherent phonon effects may further influence thermal conduction, offering a tunable platform for manipulating heat flow at the nanoscale.

NANOTHERMOMETRY

The ideal nanothermometer

The diverse range of approaches to tailoring nanoscale heat generation described so far demonstrates that the combination of plasmonic nanostructures and engineered substrates offers unique and precise control over heat generation at the nanoscale. However, this extreme localization introduces a major experimental challenge for accurately measuring and mapping temperatures at such small scales.

Traditional bulk measurement techniques fail to capture these nanoscale thermal fields. Advanced nanothermometry techniques are needed to resolve local temperatures and thermal gradients with the necessary spatial and temporal resolution, with potential impact on a wide range of technologies, from nanoscale photocatalysis to quantum computing and heat management in electronics. We identify five key criteria that nanothermometry techniques should ideally meet to accurately measure localized, time-dependent heat generation at the nanoscale: (i) sub-nm spatial resolution, (ii) fs–μs temporal resolution, (iii) sub-K thermal resolution, (iv) broad experimental compatibility and simplicity, and (v) cost-effectiveness (Fig. 5). Here, we define resolution as how spatially, temporally, and thermally close two distinct objects or events can be so that the technique can distinguish them. In this section, we will examine several micro- and nanothermometry techniques in relation to these criteria.

Currently available nanothermometry techniques

Infrared (IR) thermal imaging is one of the most conventional microscale techniques and employs IR-thermal cameras to measure the black-body radiation from a surface. The intensity recorded and displayed by each pixel corresponds to the local radiance, which is significantly influenced by the material emissivity.¹¹⁹ IR cameras are user-friendly and cost-effective lab equipment but only offer diffraction-limited resolution. This means that with the typical wavelength range of about 3–12 μm, a spatial resolution around 5–10 μm can be achieved.^{120–122} These cameras are additionally sensitive to emissivity variations, potentially creating measurement artifacts that must be carefully considered.^{123–125} Nevertheless, IR thermal imaging allows for effective microscale mapping of temperatures.

Charge-coupled device (CCD) cameras fitted with a diffraction grating are also employed for thermal imaging via quadriwave shearing interferometry (TIQSI).²⁵ Here, temperature changes are monitored by detecting the thermally induced variations in the refractive index of the medium surrounding the nanostructures,

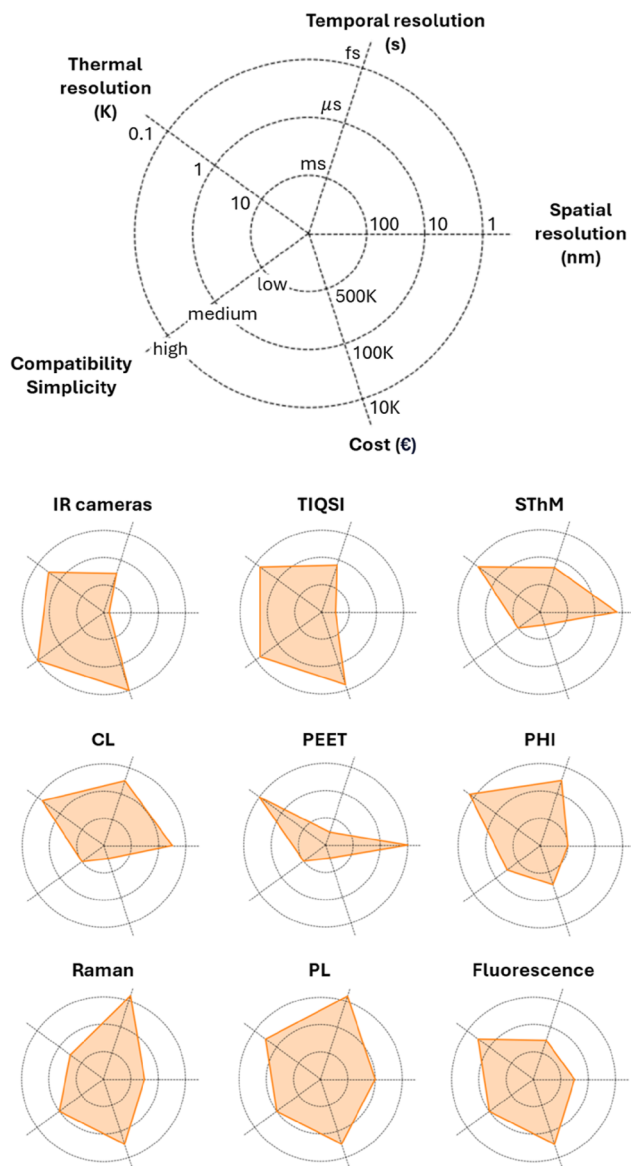


FIG. 5. Radar charts highlighting the performance of different nanothermometry techniques in the framework of the five ideal criteria. The reported charts are the state-of-the-art examples subject to individual and non-exhaustive scoring.

which manifest as optical path differences in the wavefront of transmitted light.^{126,127} The simplicity of this setup (a grating and a camera) makes it a cost-effective and straightforward method to observe plasmonic induced heating in nanostructures. The timescale for this technique is only limited by the camera's frame rate (usually ~ 10 ms), with sub-K thermal and optically diffraction-limited spatial resolution (~ 200 nm), hence much better than for IR cameras.^{15,25,40}

Thermal imaging at the nanoscale can be achieved through scanning probe microscopy (SPM)^{128,129} techniques. The most prominent example is scanning thermal microscopy (SThM), which

utilizes a built-in thermal sensor integrated onto a scanning tunneling microscope (STM) or atomic force microscope (AFM) probe. STM-based SThM detects temperature variations through changes in the tunneling current, which is influenced by the local thermal expansion, or through thermally activated tunneling processes. AFM-based SThM typically measures temperature via a thermoresistive sensor embedded in the probe tip, responding directly to nanoscale thermal gradients. Other techniques that do not rely on a built-in thermal sensor include scanning tunneling thermometry, thermal expansion thermometry, and near-field scanning optical thermometry. These techniques typically offer ~ 10 mK temperature resolution¹³⁰ and spatial resolutions well below 100 nm.^{131,132} SThM can even achieve sub-10 nm resolution, as shown in the measurement of the self-heating of nanoscale metal interconnects [Fig. 6(a)].¹³³ As a potential drawback, SThM is inherently slow, as it is limited by the tip scanning rate (generally has a ms–s time resolution), and the invasive tip–sample interaction and complex calibration make it ill-suited for ultrafast measurements. In addition, SThM is a surface-based technique, hence not capable of probing temperature gradients inside 3D samples. Nevertheless, the ultimate temporal resolving power of state-of-the-art SThM is set by the thermal time constant of the microfabricated probe itself. In practice, resistive SThM tips exhibit a thermal time constant of ~ 300 μ s.¹³⁴

Cathodoluminescence (CL) spectroscopy was proposed as another technique to map nanoscale temperature gradients.¹³⁵ CL occurs when photons are emitted as a highly energetic electron beam interacts with a material. The electron beam acts as a broadband excitation of electronic transitions, which means that the emitted CL signal is similar to the one obtained under optical excitation, i.e., displays both the luminescence peaks and the LSPRs in plasmonic materials. The local temperature can be determined by monitoring the spectral position of the CL band, provided that the optical transitions of the materials shift with temperature. This link between the temperature and the emitted spectrum was demonstrated in the measurements on semiconductor GaN nanowires with diameters in the 200–300 nm range [Fig. 6(b)].¹³⁵ Cathodoluminescence requires the use of an electron beam and has, therefore, typically been employed in scanning electron microscopes (SEMs). The incorporation of CL in a transmission electron microscope (TEM) was also proposed to further push the spatial resolution, ideally below 20 nm.¹³⁹ Despite the exciting spatial resolution, both the SEM and TEM are very expensive pieces of lab equipment and operate under vacuum,¹⁴⁰ limiting their use in practical applications. Furthermore, the potential e-beam self-heating at high currents and the need for a specialized CL detector can further increase complexity and cost.¹³⁵ Nevertheless, CL represents an interesting solution, as the temporal resolution of this technique can be pushed to the nanosecond timescale by modulating the electron beam intensity and using fast detectors.¹⁴¹

Another TEM-based method particularly suited to the characterization of heating in plasmonic nanoparticles is the measurement of plasmonic peak shifts via electron energy-loss spectroscopy (EELS). This technique is known as plasmon energy expansion thermometry (PEET).^{142,143} PEET can be used for nanothermometry by tracking temperature-induced changes in electron density and thermal expansion of nanostructures, both of which affect the EELS spectrum, enabling temperature evaluation. With such a technique, a spatial resolution of ~ 3 nm was reported,¹⁴⁴ and

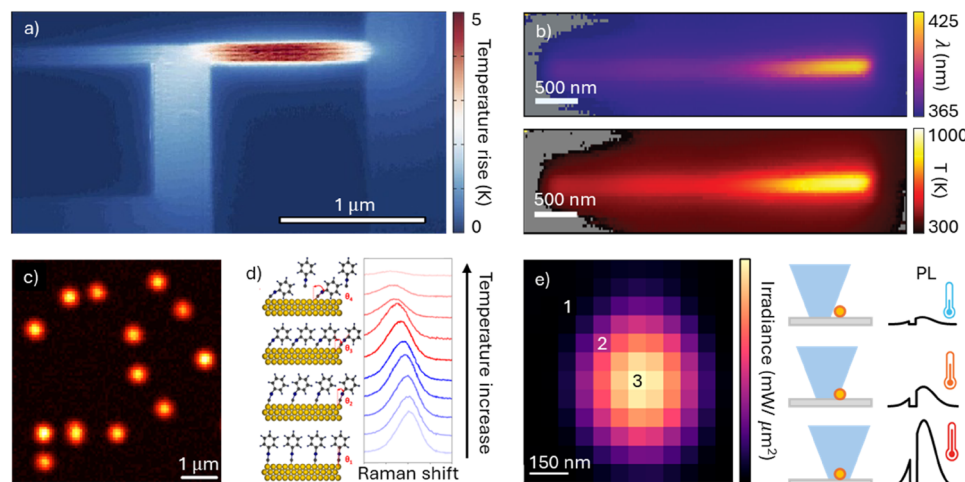


FIG. 6. (a) Self-heating of a nanoscale metal interconnect structure measured by scanning thermal microscopy (SThM). Reprinted with permission from Menges *et al.*, Nat. Commun. **7**(1), 10874 (2016).¹³³ Copyright 2016 Springer Nature. (b) Nanoscale thermometry cathodoluminescence (CL) measurement technique. Peak CL wavelength was extracted with an electron beam current of 210 nA, and the temperature map was measured by focusing the beam at each pixel. Reprinted (adapted) with permission from Mauser *et al.*, ACS Nano **15**, 11385 (2021).¹³⁵ Copyright 2023 American Chemical Society. (c) 2D photothermal heterodyne images of an area of $6 \times 6 \mu\text{m}^2$ containing individual 10 nm gold NPs. Reprinted Fig. 3(a) with permission from Berciaud *et al.*, Phys. Rev. B **73**(4), 045424 (2006).¹³⁶ Copyright 2006 American Physical Society. (d) Temperature sensing using phenyl isocyanide molecules adsorbed on an Au NP surface, with a schematic of temperature-dependent SERS measurement and temperature-induced orientation change of PIC molecules. Reprinted with permission from Hu *et al.*, J. Am. Chem. Soc. **140**(42), 13680–13686 (2018).¹³⁷ Copyright 2018 American Chemical Society. (e) Color-coded confocal image of an 80 nm Au NP colored in terms of irradiance with a schematic illustration of the heating of a NP during scanning, along with example PL spectra from pixels 1 to 3. Reprinted with permission from Barella *et al.*, ACS Nano **15**(2), 2458–2467 (2021).¹³⁸ Copyright 2021 American Chemical Society.

temperature uncertainties as low as ~ 0.2 K, depending on the quality of the EELS spectrum and the thermal expansion model used, could be achieved.¹⁴³ Nevertheless, for a highly spatially accurate temperature map, the required acquisition time is long, potentially taking 0.1 seconds per pixel.¹⁴⁵

Another interesting nanothermometry method is photothermal heterodyne imaging (PHI), a pump-probe microscopy technique in which an intensity-modulated heating beam creates a microscopic thermal lens, i.e., a local refractive-index change, around a nano-absorber [Fig. 6(c)].¹³⁶ By calibrating the PHI signal against known temperature rises and modeling heat diffusion, the heterodyne-detected refractive index changes serve as an all-optical nanothermometer, allowing the mapping of nanoscale temperature gradients in plasmonic assemblies.¹⁴⁶ Recently, it was demonstrated that the sub-microsecond temporal resolution could be pushed down to achieve nanosecond-resolved imaging of photothermal dynamics.¹⁴⁷ A related but distinct technique, heterodyne picosecond thermoreflectance (HPTR), utilizes two femtosecond lasers to achieve sub-picosecond temporal resolution and ~ 500 nm lateral resolution with <0.1 K thermal resolution.¹⁴⁸ Nevertheless, it demands a sophisticated optical modulation setup that increases cost and complexity.¹³⁶ Despite it being a diffraction-limited technique, PHI proves to be a valuable tool as it effectively locates and visualizes individual sub-10 nm absorbers with nanometer-scale precision.¹⁴⁹ PHI is indeed able to detect single non-fluorescent nano-objects down to gold clusters of just 1–2 nm.¹⁵⁰ Other pump-probe thermometry techniques include time- and frequency-domain thermoreflectance (TDTR/FDTR). TDTR uses femtosecond

pump and probe pulses on a metal transducer, and it achieves temperature decay monitoring in the order of tens of nanoseconds.¹⁵¹ However, its spatial resolution is limited by its ~ 10 – $50 \mu\text{m}$ pump/probe spot sizes.¹⁵² On the other hand, FDTR operates with a sinusoidally modulated pump (kHz–MHz) and a CW probe and can achieve 10–100 ns temporal distinguishability within similar spot sizes.¹⁵³

Other far-field optical nanothermometry techniques include Raman-, fluorescence-, and luminescence-based approaches. Raman-based nanothermometry exploits the temperature dependence of the Raman shifts^{137,154} [Fig. 6(d)] or intensity ratios of Stokes to anti-Stokes peaks^{155,156} in Raman spectra. These techniques typically offer diffraction-limited spatial resolution (down to ~ 200 nm). Surface-enhanced Raman scattering (SERS) can provide spatial resolutions approaching tens of nanometers, although distinguishing features closer than this scale (~ 10 nm) remains challenging.¹⁵⁷ The temporal resolution achievable through SERS can also extend to ultrafast regimes (sub-picosecond scale).^{158,159} Their thermal resolution typically ranges around 1–5 K, depending on materials and experimental configurations.^{160,161} An additional advantage of Raman nanothermometry is that it can be combined with operando SERS spectroscopy of chemical conversion kinetics through molecule-specific vibrational signatures.^{159,162} This multiplexed information acquisition enables a level of mechanistic understanding in plasmonic catalysis that is difficult or impossible to achieve with other techniques.

Plasmonic luminescence based techniques for nanothermometry also rely on the temperature-dependent changes in

photoluminescence intensity and anti-Stokes emission from metal nanoparticles arising from hot electron dynamics and thermalized electron distributions [Fig. 6(e)].^{163–165} Anti-Stokes thermometry, particularly from gold nanoparticles, provides high thermal resolution (~ 0.1 to 1 K) and is able to detect temperature gradients in the order of ~ 10 nm, which is attributed to the strong localization of plasmonic fields.^{138,166} Temporal resolution in these methods can achieve ultrafast scales (femtoseconds to picoseconds), beneficial for probing rapid plasmon-driven heating processes.¹⁶⁷

Similar to the Raman-based methods, fluorescence-based thermometry exploits temperature-dependent parameters such as fluorescence intensity, spectral shifts, lifetime changes, and anisotropy variations.¹⁶⁸ Techniques leveraging fluorescence anisotropy¹⁶⁹ and quantum-dot-based fluorescence^{170,171} offer high sensitivity, with thermal resolutions frequently better than 0.5 K. Spatial resolution is typically diffraction-limited but can reach tens of nanometers through advanced super-resolution methods such as the ones employing nanodiamond nitrogen vacancy centers.^{172,173} The temporal resolution generally spans from microseconds to milliseconds, governed by fluorophore lifetime dynamics.^{174,175} Each of these techniques provides complementary advantages: Raman-based methods excel in temporal resolution, fluorescence-based approaches offer outstanding sensitivity and versatility, and plasmonic luminescence excels in ultrafast, highly localized thermal measurements.

To conclude, while no single technique simultaneously achieves all five ideal criteria (Fig. 5), heterodyne and pump-probe methods excel in speed and sensitivity, electron microscopy super-resolution approaches lead in spatial precision, and simpler fluorescence or Raman schemes offer accessible trade-offs for many applications. Future advancements should focus not only on refining existing methods but also on the development of hybrid or multimodal imaging approaches, meaning using a single scanner that integrates two modalities into one physical system and/or merging separately acquired images via specifically designed and optimized software. One option could be to bridge current gaps by integrating the superior spatial resolution of electron microscopy with the temporal advantages of optical methods. However, overcoming practical limitations such as tip-induced perturbation, experimental constraints (vacuum or atmospheric conditions), and equipment complexity will remain central challenges. Innovative calibration schemes and improvements in detection sensitivity could further reduce measurement uncertainty, rendering these nanothermometry techniques more broadly applicable. Ultimately, careful selection and integration of multiple complementary techniques will provide the most comprehensive understanding of nanoscale photothermal phenomena. Such an integrated approach will facilitate obtaining further insight into thermal dynamics at unprecedented resolutions, driving the development of advanced plasmonic devices and applications.

CONCLUSION AND OUTLOOK

In summary, in this review, we have discussed the generation, modeling, and measurement of plasmonic heating. In the framework of achieving fast, intense, and very localized thermal gradients, it is critical to leverage the interplay between photons, electrons, and phonons. We have highlighted how advances in plasmonic material science, nanophotonic and heterostructure design, exploitation of

electron-phonon coupling, and tuning of the thermal properties of the micro-environment can offer precise control over the photothermal efficiency and resulting nanoscale gradients. Many of these strategies remain at an early stage, with some demonstrated only through simulations (e.g., pulsed-excitation thermal hotspots) and others experimentally verified but not yet implemented in plasmonic systems (e.g., vdW metamaterials). Realizing their full potential will require integrating these emerging concepts into functional plasmonic architectures and experimentally verifying their enhanced photothermal performance.

Meanwhile, we identified the limitations of current heat-transfer modeling. While multiple-temperature models already offer decent approximations for thermal dissipation from plasmonic NPs, upon further nanoscale confinement, more advanced non-Fourier phonon models will become crucial to account for nanoscale effects such as ballistic transport and altered local properties (e.g., electron-phonon coupling and surface scattering).

Finally, we assessed the capabilities of various nanothermometry techniques to map local temperature profiles in plasmonic systems. While existing thermometries offer either high spatial or high temporal resolution, achieving both simultaneously has remained elusive. Measuring temperatures in such small-scale volumes (down to atomic-scale resolution) would be particularly exciting, as it would enable breakthroughs in understanding heat transfer, reaction dynamics, and material properties at previously inaccessible dimensions, providing unprecedented insights into fundamental nanoscale phenomena.⁴³

Ultimately, precise control and measurement of nanoscale thermal effects are not only important for the field of plasmonics but also crucial for advancing fields such as photonics, electronics, (photo)catalysis, photo-thermoelectric energy conversion, and quantum materials. These fields often involve processes where temperature gradients occur over extremely small spatial scales. Activating and measuring these gradients could unlock improvements in energy efficiency (e.g., enhancing heat dissipation in nanoscale electronics), reaction rates (e.g., enhancing catalytic activity by localized thermal hotspots), and material properties (e.g., tailoring thermal conductivity in nanostructured materials).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Cecilia Romeo: Conceptualization (equal); Formal analysis (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Andrea Baldi:** Supervision (equal);

Writing – review & editing (supporting). **Sven H. C. Askes:** Conceptualization (lead); Funding acquisition (lead); Project administration (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

Data sharing is not applicable to this article, as no new data were created or analyzed in this study.

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